Stock Price Prediction using various method

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Introduction

1.1 Introduction and Background

Stock Price Prediction is always an interests topic to everyone in the market, particularly for computer scientists, data scientists and financial computing professions.

However, the Effective market hypothesis states that it is not possible to predict the stock and market price as the market price is efficient and already reflects all available information about the stock.

If the effective market hypothesis is true, investors is waste their time and effort to predict the stock price because any methods trying to stock price will fail and it is not possible to earn any profit from price prediction.

That means any attempt for predicting price, including fundamental analysis, technical analysis, stock news analysis, machine learning, artificial neural network, both by human and computer, are no hope to predict the price and earn any profit from prediction.

In this paper, we will first study review various of existing stock prediction strategy and technology, then we will be trying to challenge this hypothesis by building a stock prediction model and try to predict the Tracker Fund of Hong Kong (2800.HK)

We choose Tracker Funder of Hong Kong for analysis because this stock is composite of many Hong Kong shares and can reflect the Hong Kong overall stock market. We would say 2800.hk is similar to Hang Seng index, which is perceived as a one of the best stock market indicator in Hong Kong. Unlike Hang Seng Index, 2800.HK is not just a stock index indicator, but a real stock that can allow investors to invest and make profit (or loss) much easier and much more direct than Hang Seng Index. Therefore, we choose 2800.HK for investigation in this paper.

If the market is effective, we should not be able to make any prediction better than random guess. But if the market is not effective, it is possible for us to predict the price and may be able to earn profit.

2. Literature Review

2.1

2.2 A LSTM-based method for stock returns prediction: A case study of China stock market [2]

In this paper, the historical stock data for stock market in Shanghai and Shenzhen, consisting of 3049 stocks from 2013 to 2015, were acquired to analyze the stock price of China. LSTM was developed as a machine learning model to analyze sequence data. Each sequence has 30 consecutive daily stock data of each stock. 900000 sequences from 2013/07/01 to 2014/11/12 were the training data while 311361 sequences of the remaining date were validation data.

The stock returns were categorized in terms of earning rate, which were divided into seven groups/ labels as [,-1.5], [-1.5,-0.5], [-0.5, 0.4], [0.4, 1.4], [1.4, 2.5], [2.5, 4.3], [4.3,].

Various models were developed with the accuracy as follows:

- **Compared Methods:** **Random**: Prediction by chance; **M1**: LSTM with closing price and trade volume as learning feature; **M2**: M1 with normalization; **M3**: M2 with extra 3 features added (High, Low, Open); **M4**: M3 with extra 5 features added (High, Low, Open, Close, Volume of Shanghai Securities Composite Index (SSE Index (000001.SS)); **M5**: only stocks of SSE ETF180 Index used for training and validation. We summarized results of different methods in **Table 1**.

Table 1. Results of Stock Return Prediction

Methods	Features	Accuracy
Random	NA	14.3%
M1	Close, Volume	15.6%
M2	Close, Volume	19.2%
M3	High, Low, Open, Close, Volume	20.1%
M4	SSE Index (Close, High, Low, Open, Volume), High, Low, Open, Close, Volume	24.1%
M5	SSE Index (Close, High, Low, Open, Volume), High, Low, Open, Close, Volume	27.2%

From the above table, it shows that the inclusion of more features will yield better accuracy of the stock prediction, especially including SSE Index, implying that the overall market trend/ performance has effects on the stock returns. Furthermore, from the model of M5, which used only stocks of SSE ETF180 Index as training and validation, has further improved the accuracy of the model. It shows that various types of stocks set will affect the accuracy of the prediction. For instance, if we put highly correlated stock sets as features, we will predict the stock returns more accurately since they have a relatively obvious relationship.

The accuracy of the models predicting stock returns is around 20 to 30%, which seems that the performance is low. However, the prediction of stock return is not an easy task since many factors affected the stock returns, including socio-economic development, global economic situation, economic cycle etc. One of the approaches to enhance accuracy is to include the international market indexes, breaking of financial news, price of bulk commodity etc. as the features for the model. The international market indexes to certain extent reflects the reality of the China stock market because China has globally traded with other countries. The breaking of financial news can indicate the points which have sharp rise/ fall. The Price of bulk commodity can indicate the economic situations of the region/ country to predict the stock market effectively because some commodities like crude oil's price affects the investors' motivation on decisions, thus the stock market.

2.3 Comprehensive Study of Market Prediction from Efficient Market [3]

In this section, we will review a study of stock prediction paper "Comprehensive Study of Market Prediction from Efficient Market published in Computational Economics. The paper first discusses the Stock predictability. Stock predictability is always a hot topic has been argued for long time. Especially predicting stock market using financial time series, some scholar belief that price can be predict using time series but most of scholar are skeptical. In 1965, Fama and Samuelson use the term efficient market and state that if the market is efficient that it cannot be predict. The random walk hypothesis state that stock market prices are evolved by random walk, which means price change are random, and therefore price cannot be predicted.

There are many scholars has studied the market historical data and the evidence of efficient market hypothesis, and found that there is evidence to reject the random walk hypothesis in the weekly data of some historical stock market. Researcher also found that stock may be predict to some extends. Researcher also found that, using the non-linear approach to study financial time serious, can predict deeper and better.

In 2004, a group of researchers observe that there are unexplained volatility after each release of information after study the stock movement of standard and poor return, they develop a new term called adaptive market hypothesis to explain the efficiency in the financial market. In this hypothesis, market is not totally rational behavior but a combination of complicated psychological factors.

The paper also discusses machine learning and prediction technique. There are many machine learning techniques to study and predict the market. Besides time series analysis, there are machine learning technique to extract the availability of social media data and use machine learning to analysis them. In the paper, the researcher attempts to predict Standard and Poor (S&P) 500 Index with latest advanced deep learning method. The objective of the study is to determinate whether effective market hypothesis is valid or not.

Before prediction, data selection and feature selection are very important. Stock prediction error will occur if there are problem in data generating process and feature selection process. In the research paper, the author discussed popular method like Principal Component Analysis, Random Walk, Deep Neural Network (like convolutional neural network and recurrent neural network).

In the research paper, the author uses Artificial Neural Network Model to predict the stock, which is a classical machine learning computational model inspired by human neural network system. However, studies also reveal that Artificial neural network are not necessary perform better than conventional statistical method when predicting time series. But it is generally believed that Artificial neural network generally performs better than others method. To be precise, the author uses PCA-LSTM, RF-LSTM, Deep LSTM and Wavelet-Deep LSTM model, which is a four deep artificial neural network model to predict the stock.

Before building the model, the author will first use other artificial intelligence technique, like random forest, wavelet, principal component analysis to do some data preprocessing and feature selection, these data preprocessing can help us to prevent overfitting problem, reduce data dimensionality, and reduce data complexity. Then we need to ensure data stationary, the author use unit root test to study. However, the author points out that some research found that non-stationary time series sometimes may work better but no evidence support this claim.

The author analysis S&P 500 from September 2000 to February 2021 and data are extracted from Yahoo Finance. The author uses Sep 1 2000 to Oct 3 2018 as training data, Oct 4 2018 to 11 Sep 2020 as validation data and Sep 12 2020 to 25 February 2021 as test data.

There are many Parameters and hyperparameters need to consideration on building model. During learning, the author found that tanh activation tends to better than ReLU function in terms of avoiding overfitting issues. Besides, adding too much layers to the artificial neural network may cause primary layer weight and stop improving. The author uses Adam optimizer as gradient descent optimizers. The author studies a lot and found that a batch size of 22 and decay rate of 0.007 is good for learning. The author built 4 experiment, one use Principles Component Analysis to reduce data dimension, one use Random Forest as feature selection, one use a high number of parameters, and one use wavelet transform.

To evaluate which perform the best, the author is not only use metrics that measure the difference between actual and prediction value such as MAE, MSE, MAPE and R Square, but also use CSP, CDCP, and Diebold-Mariano statistical test to evaluate the accuracy, which is better for measure profitability prediction.

The author conducts 5 experiments, the first experiment known as PCA and deep learning, which use Principal Component Analysis for pre-extract knowledge from high dimensional data to lower dimensional. The model has 4 hidden LSTM layers.

The network is utilized with dropout and batch normalization layers to reduce the risk of overfitting. The experiment has 92,250 parameters. The second experiment is to use Random Forest as feature selection. Again, this model has 4 hidden layers, also utilized with dropout and batch normalization layers. It contain 108,842 parameters. The third experiment does not adopt feature selection or dimensional reduction algorithm, make it higher parameters, it uses ReLU as activation function. It has 158,810 parameters. The fourth experiment use wavelet transform before fitting the neural network, and uses a similar method as experiment 3, it contains 205,420 parameters. The benchmark method has no feature engineering and no denoising, with 4 densely connected neural networks implemented.

Finally, the experiment result is follows

Validation and test data results obtained from the compared experiments

Experiment	MAE	MSE	MAPE	CSP	CDCP
	Validation data				
PCA and deep learning	0.00990	0.00028	1.637	0.555	0.725
RF and deep learning	0.00997	0.00028	1.312	0.567	0.721
Deep learning	0.00972	0.00028	1.429	0.581	0.737
Wavelet and deep learning	0.00971	0.00025	1.502	0.575	0.725
Benchmark method	0.00997	0.00028	1.406	0.567	0.721
	Test data				
PCA and deep learning	0.00757	0.000106	1.169	0.552	0.719
RF and deep learning	0.00783	0.0001042	1.089	0.543	0.719
Deep learning	0.00790	0.0001044	1.375	0.543	0.736
Wavelet and deep learning	0.00810	0.000110	1.478	0.517	0.692
Benchmark method	0.00783	0.000104	1.186	0.543	0.719

As we can see from the above table, the error rate of validation data are higher than test data, meaning that there may have potential over-fit in this model. However, for both validation data and test data, the mean absolute error (MAE) is lower than 0.01, which is considerably very low, meaning the prediction rate is quite high. We can found that LSTM is a good way for us to predict the stock market, no matter what feature selection method use. Among all methods, the PCA and Deep learning seems to perform the best for test data and wavelet perform best for validation data.

Residual analysis

Residual diagnostics	Mean (Bias error)	Standard Deviation	Maximum loss
PCA and deep learning	−0.0011	0.01028	0.02117
RF and deep learning	0.0004	0.01026	0.02139
Deep learning	−0.0001	0.01027	0.02110
Wavelet and deep learning	0.0010	0.01051	0.02120
Benchmark method	0.0000	0.01027	0.02142

For the residual analysis, both bias error, standard deviation and maximum loss is seeming quite low. The bias error is generally lower than 0.01. The loss is lower than 0.02, which is good.

Results from Diebold Mariano test on four the experiments conducted

D-M test/Wilcoxon signed rank test	PCA and deep learning	RF and deep learning	Wavelet and deep learning	Deep learning	Benchmark method
PCA and Deep Learning		***P* value=0.000** **Wilcoxon stats=1032.0**	***P* value=0.000** **Wilcoxon stats=1153.0**	*P* value=0.908 Wilcoxon stats=3237.0	***P* value=0.002** **Wilcoxon stats=2203.0**
RF and Deep Learning	*P* value=0.352 D-M stats=0.934		*P* value=0.543 Wilcoxon stats=3063.0	***P* value=0.000** **Wilcoxon stats=468.0**	***P* value=0.000** **Wilcoxon stats=1958.0**
Wavelet and Deep Learning	*P* value=0.106 D-M stats=0.509	*P* value=0.214 D-M stats=1.247		***P* value=0.000** **Wilcoxon stats=1292.0**	***P* value=0.000** **Wilcoxon stats=1634.0**
Deep Learning	*P* value=0.236 D-M stats=1.191	*P* value=0.467 D-M stats=0.728	*P* value=0.402 D-M stats=−0.840		*P* value=0.150 Wilcoxon stats=2769.0
Benchmark Method	*P* value=0.315 D-M stats=1.008	*P* value=0.943 D-M stats=−0.070	*P* value=0.196 D-M stats=−1.298	*P* value=0.405 D-M stats=−0.835	

Those pair of models with significant variation of error term with 99 precent confidence interval are bold faced

Finally, the author conducts Diebold-Mariano test, which is used to determine whether these forecasts are significantly different. From the result, it shows that the prediction accuracy between each model is not significant.

Wilcoxon stats, or Wilcoxon signed-rank test, is a statistical hypothesis test used to test the location of a population based on a sample of data, or to compare the locations of two populations using two matched samples. In this model, the Wilcoxon Stats shows that the PCA LSTM neural network seems perform the best in terms of knowledge extraction.

2.4 Article 4

2.5 Impact of Financial Ratios and Technical Analysis on Stock Price Prediction Using Random Forests [5]

This paper study different technical analysis method, Random Forest was found to have good prediction result. Random Forest's random factor makes the prediction more robust to noise and resistant to over training.

The study makes use of Random Forest to study 433 companies listed in the Hong Kong Stock Exchange from 2011-2014, each with a set of 63 attributes in which the majority of them were related to financial ratios per quarter, including Liquidation

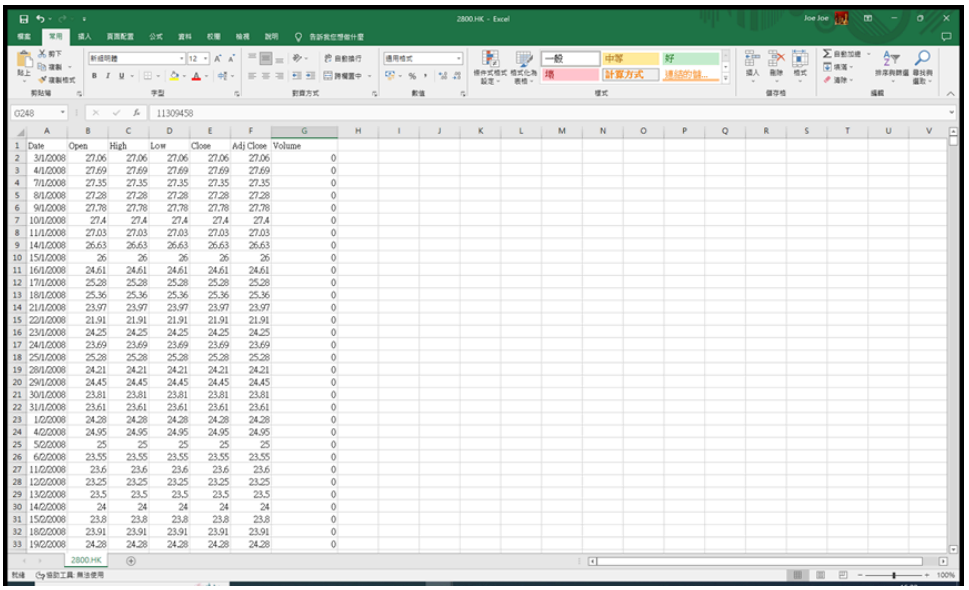
value/Market Cap, Book Asset Value/Market Cap, Net Asset Value/Total Assets etc. and their variations. the accuracy is then compared. Although the data has been normalized to have zero mean and variance of 1, the report accuracy was still misleading because of skewed class distribution when using percentage to represent the correctly classified instance. So, Area Under Curve (AUC) were introduced to make fair comparison.

It was found that in some time period, the stock price were fluctuating, the simple model was not powerful enough to take into consideration. The author implied from the result that the correct predictive model in most cases is only useful within a period of time and may be required to be updated when there is a change in the operating environment.

3. Stock Prediction - By Technical Analysis
3.1 Moving Average (By Student A)
3.2 Exponential Moving Average (By Student B)
3.3 MCAD (By Student C)
3.4 RSI (By Student D)
3.5 Bollinger Bands (By Student E)

4. Stock Prediction - By Machine Learning
4.1 Data Preprocessing

We first obtain the stock data 2800.HK from Yahoo Finance.



Date	Open	High	Low	Close	Adj Close	Volume
3/1/2008	27.06	27.06	27.06	27.06	27.06	0
4/1/2008	27.69	27.69	27.69	27.69	27.69	0
7/1/2008	27.35	27.35	27.35	27.35	27.35	0
8/1/2008	27.28	27.28	27.28	27.28	27.28	0
9/1/2008	27.78	27.78	27.78	27.78	27.78	0
10/1/2008	27.4	27.4	27.4	27.4	27.4	0
11/1/2008	27.03	27.03	27.03	27.03	27.03	0
14/1/2008	26.63	26.63	26.63	26.63	26.63	0
15/1/2008	26	26	26	26	26	0
16/1/2008	24.61	24.61	24.61	24.61	24.61	0
17/1/2008	25.28	25.28	25.28	25.28	25.28	0
18/1/2008	25.36	25.36	25.36	25.36	25.36	0
21/1/2008	23.97	23.97	23.97	23.97	23.97	0
22/1/2008	21.91	21.91	21.91	21.91	21.91	0
23/1/2008	24.25	24.25	24.25	24.25	24.25	0
24/1/2008	23.69	23.69	23.69	23.69	23.69	0
25/1/2008	25.28	25.28	25.28	25.28	25.28	0
28/1/2008	24.21	24.21	24.21	24.21	24.21	0
29/1/2008	24.45	24.45	24.45	24.45	24.45	0
30/1/2008	23.81	23.81	23.81	23.81	23.81	0
31/1/2008	23.61	23.61	23.61	23.61	23.61	0
1/2/2008	24.28	24.28	24.28	24.28	24.28	0
4/2/2008	24.95	24.95	24.95	24.95	24.95	0
5/2/2008	25	25	25	25	25	0
6/2/2008	23.55	23.55	23.55	23.55	23.55	0
11/2/2008	23.6	23.6	23.6	23.6	23.6	0
12/2/2008	23.25	23.25	23.25	23.25	23.25	0
13/2/2008	23.5	23.5	23.5	23.5	23.5	0
14/2/2008	24	24	24	24	24	0
15/2/2008	23.8	23.8	23.8	23.8	23.8	0
18/2/2008	23.91	23.91	23.91	23.91	23.91	0
19/2/2008	24.28	24.28	24.28	24.28	24.28	0

4.2 Model 1

4.2.1 Data Cleansing

Data cleansing is an important part of data analysis.

After downloading the CSV file, we notice that there are some issues for the data. First, the Yahoo Finance does not provide correct volume trading in the early stock data. For those stock data without any volume information, Yahoo Finance put it into zero value. However, it is definitely will affect the stock prediction result. There is also some null value, which may also affect the stock prediction result.

Since our objective for this dataset is to predict price, we can make some assumption to clean the data and make some adjustment to fit this purpose and enhance the prediction analysis result.

In this dataset, there are just a few columns that have null value, but this will affect prediction. For the null value, we choose the average of the previous 14 days to replace null value. This can be done by using excel formula
AVERAGE(INDIRECT(CHAR(COLUMN()+65-1)&ROW()-14):INDIRECT(CHAR(COLUMN()+65-1)&ROW()-1))

For the Date Format, we put all data format to yyyy-MM-dd to avoid confusion and ambiguity.

After data cleansing, we can save the new CSV file.

4.2.2 Data Preprocessing
After Data Cleansing, we need to add a new column "Next Price", which is the closing price of the next day. This can be done by excel formula, which is reference to the next row of the closing price.

To make a better prediction, we can also add more columns to reflect the previous day of price, say previous day 1, previous day 2, previous day 3, previous day 4, previous day 5 price. In our model, we add 5 new columns representing the previous 1 to previous 5 day of closing price respectively.

4.2.3 Introduction to Prediction Method

We have developed several models for stock analysis.

The first model is linear regression. Linear Regression is a simple but powerful machine learning technique. It is a statistical learning method, which found and modelling the relationship between an explanatory variable (known as independent variable, commonly named as x variables), and a scalar response (known as dependent variable, commonly named as y variables). The equation of linear regression is written as y= mx + c (we predict y based on x). In order to determine the regression line, we can first calculate the m and c.

Normally, in linear regression, m is equal to sum of (x - mean of x) * (y - mean of y) / sum of (x – mean of x)^2

c is computed by substituting the mean of x, the mean of y and a into y = b + ax => mean of y = b + a (mean of x) => solve for b
Based on generated linear regression formula, we can predict tomorrow price based on historical price.

We can also predict tomorrow price based on other factor that have linear correlation with the stock price such as volume, fundamental information like p/e or p/b ratio, price changes.

In excel, we can use the Excel Formula **=forecast.linear(x, know_ys, know_xs)** to generate a linear regression prediction result. Although linear regression can be done in excel prediction. We found that Orange data mining are even work better, so we choose Orange data mining for linear regression for this analysis

The second model we use for prediction is K Nearest Neighbour (KNN). KNN is a supervised learning classifier uses to make classification or prediction on the grouping of individual data point.

The third model is random Forest is a machine learning method allow us to use historical price to build a decision tree then predict tomorrow price. The algorithm is based on historical price or certain information to forming many decision trees (a forest) each providing a prediction, the overall predictions can be found by computing the mean.

The fourth model is an artificial neural network. An Artificial Neural Network (ANN) is a core concept in modern machine learning, originally inspired by the biological human brain. It is one of the most powerful mathematical and statistical models that can help us to solve many problems in many areas, such as computer vision and face recognition, predicting disease, predicting stock and housing price, predicting market, machine translation, self-driving, games, etc. It can consider as generic learning model

An ANN is based on a collection of connected units or nodes called artificial neurons. The artificial neuron receives one or more inputs and sums them to produce an output (or activation). This is inspired by our brain excitatory postsynaptic potentials, inhibitory postsynaptic potentials at neural dendrites, and neuron's action potential). Each connection can transmit a signal to other neurons, which simulate the synapses in our brain. An artificial neuron receives signals then processes them and can signal neurons connected to it.

The "signal" at a connection is a real number, and the output of each neuron is computed by some non-linear function of the sum of its inputs.
The connections are called edges. Neurons and edges typically have a weight that adjusts as learning proceeds.
The weight increases or decreases the strength of the signal at a connection.
Neurons may have a threshold such that a signal is sent only if the aggregate signal crosses that threshold.

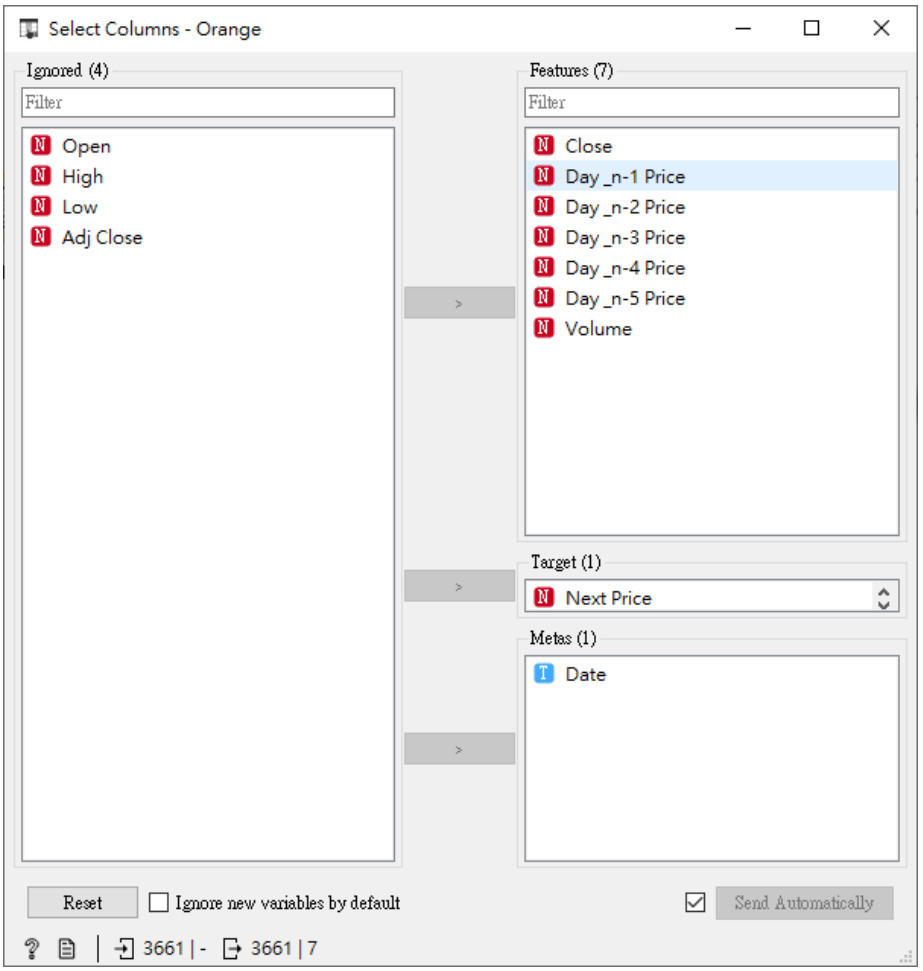
The fifth model supports vector machines (SVM). SVM is a supervised

learning model with associated learning algorithms that analyze data for classification and regression analysis. SVM is a statistical learning framework. Support vector machines are used to sort two data groups by like classification. The algorithms draw lines (hyperplanes) to separate the groups according to patterns.

4.2.4 Prediction

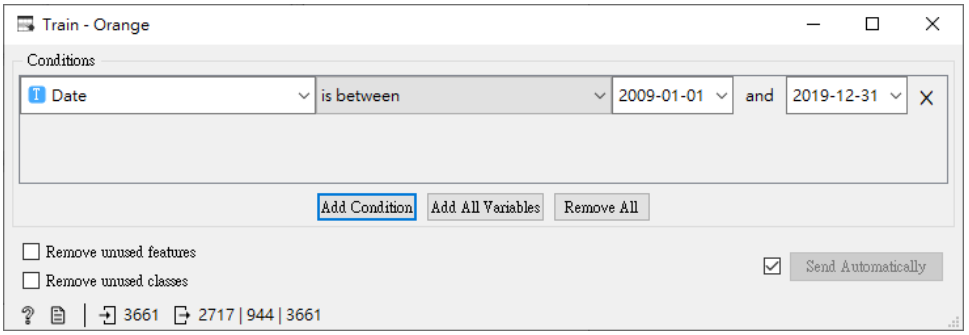
In this project, I use orange data mining to build a machine learning model based on various artificial intelligence techniques, including SVM, KNN, Neural Network and Random Forest, and Linear Regression.

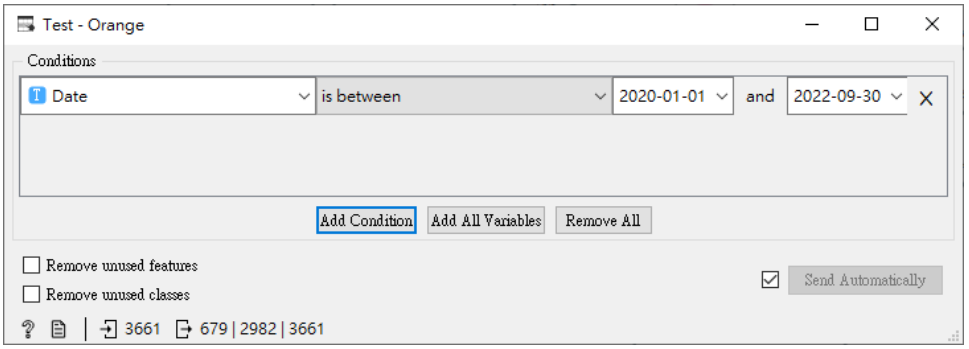
For the Feature Selection Column, I take current closing price, previous 1 closing price, previous 2 closing price, previous 3 closing price, previous 4 closing price, previous 5 closing price and volume as feature for prediction, the target is the next price for our target prediction, and date is a meta value.



In this model, I choose the dataset from 2009-01-02 to 2022-09-30 for prediction.

I would choose the date between 2008-01-10 to 2019-12-31 as training data, and the 2020-01-01 to 2022-09-30 as test data.

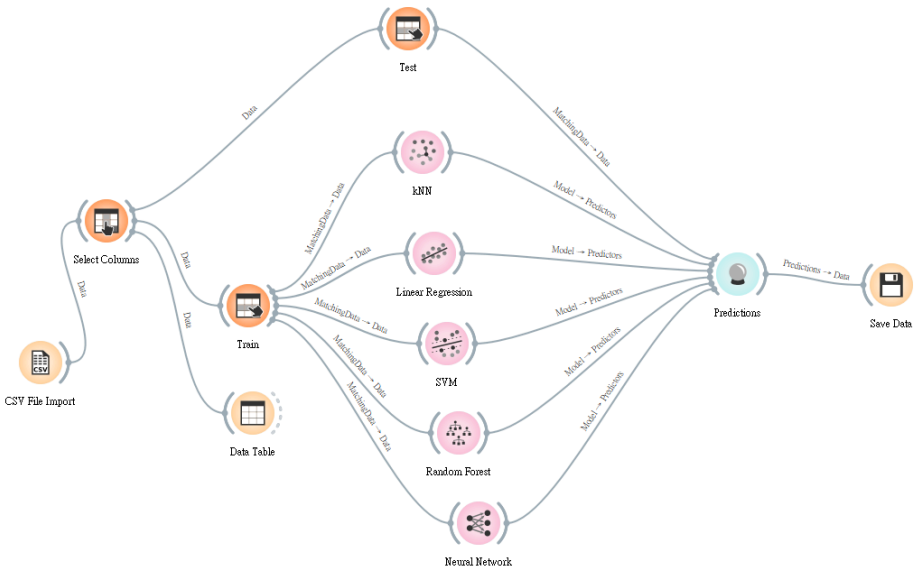




For the Neural Network, we use 100 hidden layers, with ReLU as activation function, and Adam as optimal solver.

For the KNN, we use 5 number of neighbours.

For the Random Forest, we choose 10 decision trees, to limit the growth of random forest, we will not split if the tree are less than 5.



Because in this model, we are using the data from 2009 to 2019 to predict the stock data from 2020 to 2022. From 2020 to 2022, we have face many challenging and unexpected situtation like Global COVID-19 outbreak, COVID-19 Lockdown, sanaction from foreign countries, tension between China and US, Russia invasion to Ukraine, and these unexpected social and hygenie events can certainly affect the price. The prediction is quite challenging, I don't expect the result is good.

Now, Let's take a look of the Prediction Result is follows

Model	MSE	RMSE	MAE	R2
Random Forest	0.157	0.396	0.302	0.982
kNN	9.270	3.045	2.528	-0.090
SVM	1.885	1.373	1.052	0.778
Neural Network	2.171	1.473	0.503	0.745
Linear Regression	0.137	0.370	0.273	0.984

679 | 679 | 5×679

In this five model, we can found that the Linear Regression and Random Forest predict the best in terms of mean absolute error, with the mean absolute error of only 0.273 and 0.302 respectively.
With the average stock price is around 25.19, the mean absolute error of 0.273 means the error rate is less than 1%, which seems quite good.

Surprisingly, the neural network is not really perform better than linear regression and random forest. Although the term neural network seems work the best, but actually sometime simple machine learning method like linear regression can even predict better.

KNN predict the worst, it's mean absolute error is nearly 2.528, which not good.

Although prediction seems good, whether it can earn money are another story, we will further be evaluated by simulated trading on section 5.1

Besides studying the test data, we will also put the training data prediction and merge it together for further analysis.

4.3 Model 2

In this model, LSTM (% change) will be used to predict the stock price. The close price of the stock is used to compute the percentage change of the stock price. The input of the model is the sequence of the percentage changes of the stock prices of the previous 40 days. The output is the percentage change of the stock prices of the next

day. Another model with the sequence of the percentage changes of the stock prices of the previous 60 days has also developed to make some comparisons.

Below is the model setting.

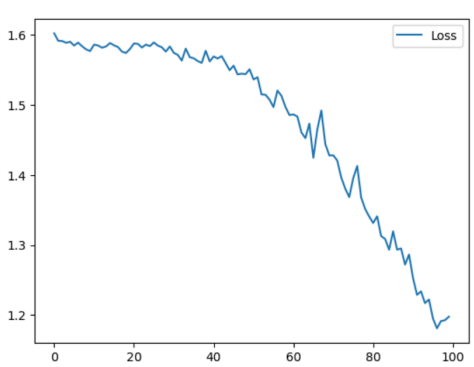
```
model.compile(loss='mean_squared_error', optimizer='adam')
```

```
history=model.fit(X_train, y_train, epochs=100, batch_size=64)
```

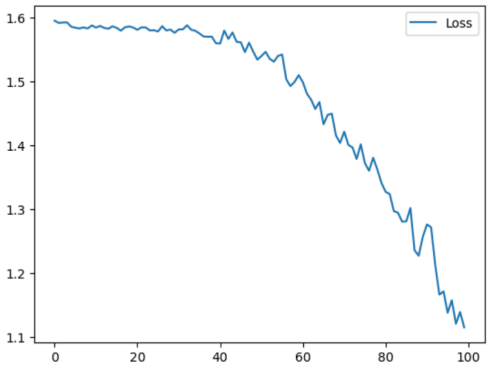
Below is the structure of LSTM.

```
model = Sequential()
model.add(LSTM(units=80, return_sequences=True, input_shape=(X_train.shape[1], 1)))
model.add(Dropout(0.2))
model.add(LSTM(units=80,return_sequences=True))
model.add(Dropout(0.2))
model.add(LSTM(units=80))
model.add(Dropout(0.2))
model.add(Dense(units=1))
```

Below are the training loss of the model previous 40 days and 60 days.



(previous 40 days)



(previous 60 days)

The prediction price by LSTM models and real price

Below is the prediction error.

	Mean Absolute Error	Mean Square Error	Coefficient of Determination
Previous 40 days	0.298076	0.158869	0.981178
Previous 60 days	0.30098	0.164034	0.980567

From the above figure, roughly similar prediction results can be obtained in the previous 40 days' and 60 days' models. Both predictions are very close to the real stock price. The previous 40 days' model performed a little bit better than the previous 60 days' model.

4.4 Model 3 (By)
4.5 Model 4 (By)

4.6 Model 5 (By)

5. Evaluation
5.1 Evaluation by Automated Trading Simulation
Although we have presented many stock prediction models and evaluated by many indicators such as mean absolute value, it is still meaningless to investors because we don't know whether this figure actually bring us to make a profit. It is possible that the Mean Absolute Value is low, and seems 99% prediction correct, but finally turn out totally loss without making any profit. On the other hands, it is possible for us to make profit with bad prediction and a good trading strategy. Making stock prediction without making any trading simulation is very dangerous and risky to any investors. Therefore, the only way to validate the effectiveness of the prediction model is to make an simulate automated trading based on the prediction and our trading strategy.

We have developed an automated trading script for our evaluation. The trading script is a simulated trading only and is written using Excel VBA Macro.

In this paper, we used various trading strategies to evaluate the model. First, we assume we have $500,000 cash and 10,000 stock of 2800.HK. The minimum saving is $50,000 which means we will not buy any stock if we cannot have $50,000 cash after finishing the transaction. The transaction fee is 1% of the transaction amount and the minimum transaction fee is 100. The trading strategies is mixed the Bollinger Bands Method and Prediction Model. We buy the stock if the current stock is lower than the lower bands (i.e. 2 times of the standard deviation of the 20-day moving average) and the current stock price is lower than the Prediction Price. We sell the stock if the current stock is higher than the upper bands (i.e. 2 times of the standard deviation of the 20-day moving average) and the current stock price is higher than the Prediction Price. If we reach the buying signal, we will buy 20% of our cash. If we reach the selling signal, we will sell 20% of our cash.

The Trading Strategy can be present as follows

Initial Money	HK$500,000.00
Initial Number of Stock	10,000
Transaction Charge	1%
Minimum Transaction Charge	HK$100.00
Buying Signal	1) Prediction Price is Higher than Current Stock Price 2) Current Stock Price is lower than the Lower Bands (i.e. Lower than 2 Standard Deviation of the 20 days moving average)
Selling Signal	1) Prediction Price is Lower than Current Stock Price 2) Current Stock Price is higher the Upper Bands (i.e. Higher than 2 Standard Deviation of the 20 days moving average)
Buy Amount	20% of our cash
Sell Amount	20% of our stocks

The Trading Strategy and Evaluation Result are presented as follows:

After study various of model, we found that the accuracy of stock prediction may not necessarily guarantee to the profitability. Whether we can make money or not depends not only on our ability to forecast the stock price, but also the trading strategy. Trading strategies are more important than prediction. In this model, although the mean absolute error of KNN and SVM are higher than neural network and random forest, they made the most money under our purposed trading strategy. Random Forest, which makes the least mean absolute error value, produces the least profitability.

	Initial	**No Trade**	**Trade (Model 1)**	**Trade (Model 2)**
Date	**2009-01-02**	**2022-09-30**	**2022-09-30**	**2022-09-30**
Money	**$500,000.00**	**$500,000.00**	**$73,943.52**	**$271,284.41**
Number of Stock	**10,000**	**10,000**	**38,000**	**37,000**

Closing Stock Price at that day	**$15.18**	**$17.89**	**$17.89**	**$17.89**
Prediction Method	**-**	**-**	**Linear Regression**	**Neural Network**
Mean Absolute Error	**-**	**-**	**0.25**	**1.89**
Total Assets	**$651,800.00**	**$678899.99**	**$753,753.48**	**HK$933,214.37**

	Trade (Model 3)	**Trade (Model 4)**	**Trade (Model 5)**
Date	**2022-09-30**	**2022-09-30**	**2022-09-30**
Money	**$72,166.61**	**$165,722**	**$73508.66**
Number of Stock	**42,000**	**43,000**	**35,000**
Closing Stock Price at that day	**$17.89**	**$17.89**	**$17.89**
Prediction Method	**KNN**	**Random Forest**	**SVM**
Mean Absolute Error	**0.67**	**0.76**	**0.19**
Total Assets	**$823,546.57**	**$935,041.96**	**$699,658.62**

In the above model, it clearly shows that all 5 models can earn profit, with Random Forest and Neural Networks earning the most. Although the mean absolute error of SVM is the lowest, it also earns lowest profit. It seems that there is no very very strong relationship between prediction rate and profitability, the key of profit is the mixture of trading strategy with the prediction, with trading strategy playing a very important and critical role.

However, it may give us an illusion for profit if we include both train data and test data for prediction, what about if we just predict for the test case only?

Here is our result for the test data

	Initial	No Trade	Trade (Model 1)	Trade (Model 2)
Date	**2020-01-02**	**2022-09-30**	**2022-09-30**	**2022-09-30**
Money	**$500,000.00**	**$500,000.00**	**$68,672.36**	**$141,217.69**
Number of Stock	**10,000**	**10,000**	**27,000**	**28,000**
Closing Stock Price at that day	**$28.70**	**$17.89**	**$17.89**	**$17.89**
Prediction Method	**-**	**-**	**Linear Regression**	**Neural Network**
Mean Absolute Error	**-**	**-**	**0.273**	**0.503**
Total Assets	**$787,001.01**	**$678899.99**	**$551,702.36**	**HK$606,357.66**

	Trade (Model 3)	Trade (Model 4)	Trade (Model 5)
Date	**2022-09-30**	**2022-09-30**	**2022-09-30**
Money	**$73,934.5**	**$105,918.19**	**$66,659.47**
Number of Stock	**28,000**	**28,000**	**28,000**

Closing Stock Price at that day	**$17.89**	**$17.89**	**$17.89**
Prediction Method	**KNN**	**Random Forest**	**SVM**
Mean Absolute Error	**2.528**	**0.308**	**0.778**
Total Assets	**$564,854.47**	**$608,838.16**	**$567,579.44**

From the result we can found that, if we only use test data, it turns from profit to loss. All 5 models lost money. This may reflect the truth of our investment. The test data can eliminate overfit problem and help us to understand the true predictability of the model.

6. Summary and Conclusion

There is evidence to reject the effective market hypothesis, it is possible to make some prediction of the stock price by using machine learning, or other analytical tools. However, it does not guarantee you can make a profit even if you make a good prediction. In fact, trading strategy is more important when we talk about profitability. For an investor, instead of focusing on stock forecasts, we should also put more effort into researching the trading strategy.

Simulated automated trading is critical, it is dangerous and risky for any investor who just invests based on the forecast model without making any simulated trading automation based on the forecast model.

7. References

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8. Contributions

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Section 1.1 Introduction and Background
Section 2.1 Literature Review - Comprehensive Study of Market Prediction from Efficient Market
Section 4.1 Data Preprocessing
Section 4.2 Model 1
Section 5.1 Evaluation by Automated Trading Simulation
Automated Trading Simulation Macro VBA Programming